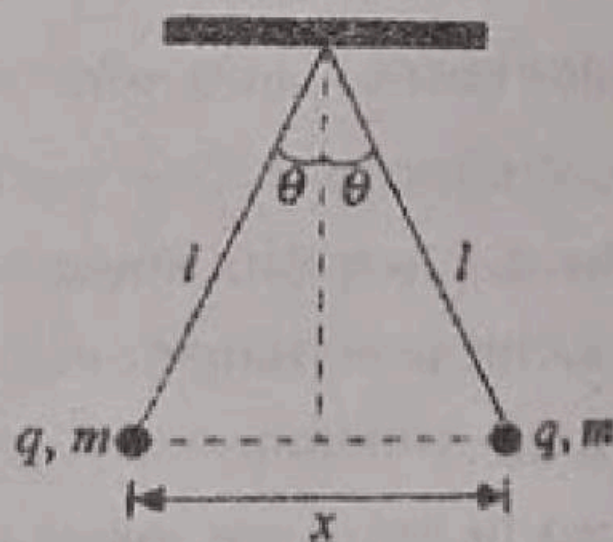


- c) Mercury boils at 357°C . However, mercury thermometers are made such that they can measure temperature up to 500°C . Explain how this is possible. [1]
- d) Suppose you come across old scientific notes that describe a temperature scale called Z on which the boiling point of water is 65.0°Z and the freezing point is -14.0°Z . To what temperature on Fahrenheit scale would a temperature of $T = -98.0^{\circ}\text{Z}$ correspond? Assume that Z scale is linear, that is, size of a Z degree is the same everywhere on the Z-scale. [3]
- a) Explain modern theory of electrification? [1]
- b) Define 1 Coulomb of charge on the basis of Coulomb's law. [1]
- c) When a glass rod is rubbed with a silk cloth, charges appear on both. A similar phenomenon is observed with many other pairs of bodies. Explain how this observation is consistent with the law of conservation of charge. [2]
- d) Two small spheres each of mass m and positive charge q , hang from light threads of lengths l . Each thread makes an angle of θ with the vertical as shown in the figure.



- (i) On the diagram draw and label all the forces on spheres. [1]
- (ii) Develop an expression for the charge q in terms of m , l , θ and g and the Coulomb's law constant. [3]

Attempt all the questions:

GROUP 'B' [8×5=40]

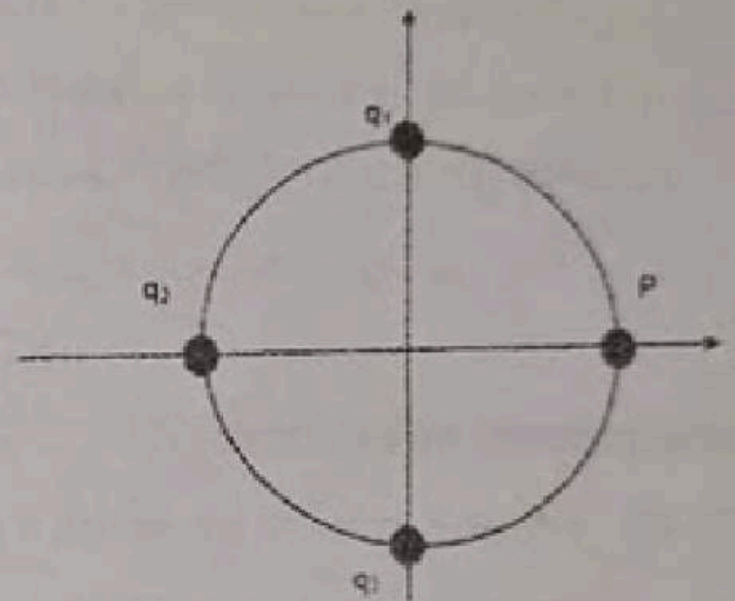
Short answer questions:

12. a) What are dimensional and non-dimensional variable? Give examples of each. [2]
- b) Check the correctness of the physical relation $Y = \frac{M g L}{\pi r^2 l}$ where Y = Young's modulus Mg is weight applied on wire, L is length of wire, l is increase in length of wire, and r is radius of wire. [2]
- c) How many times the units of energy be increased if each of the units of mass, length and time is increased by two times. [1]
13. a) Two people are pulling a heavy cart from the same point but in slightly different directions. One pulls with a force of magnitude A , the other with force B , and the angle between their ropes is θ . The cart doesn't turn—it moves straight in a direction that results from both pulls happening simultaneously. Using the rule that combines two vectors acting from the same point at an angle, derive an expression for the total force acting on the cart (its magnitude and direction). [3]
- b) The two forces are given by: $\vec{F}_1 = 500 \text{ N, east}$ and $\vec{F}_2 = 600 \text{ N, north}$. Draw a vector diagram to subtract $\vec{F}_2$ from $\vec{F}_1$. Find the magnitude and direction. [2]
14. a) A scalar may be a pure number or a physical quantity. What is the effect of multiplication of a vector by a scalar? [2]
- b) Give the geometrical interpretation of dot product of two vectors. [1]

- c) If $\vec{A} = 4\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{B} = -2\hat{i} - 5\hat{j} + 7\hat{k}$, find the angle between $\vec{A}$ and $\vec{B}$. [2]

15. a) The expression of electric field E due to a point charge at any point near to it is defined by $\vec{E} = \lim_{q \rightarrow 0} \frac{\vec{F}}{q}$, where q is the test charge and F is the force acting on it. What is the significance of $\lim_{q \rightarrow 0}$ in this expression? [1]
- b. Why two electric field lines cannot intersect with each other? [1]

- c) Consider three equal charges q_1, q_2 , and $q_3 = q = +1\mu\text{C}$ located at three different points on a circle of radius 1m, as shown in the figure. Calculate the total electric field at point P due to this distribution of charges. [3]



16. a) What are induced and inducing charges. How will you charge a conductor positively by the method of induction. Explain with clear diagram. [1+2]
- b) A small ball of paper has a mass of 9×10^{-5} kg and carries a charge of $5\mu\text{C}$. When it is held over another charged ball of paper at a distance of 2 cm above it, the two balls stay in equilibrium, what is the charge on the second ball? [2]
17. a) What are electric field lines? [1]
- b) If one field line represents 1.0×10^{-9} C of charge, determine the number of field lines drawn for a charge of $2\mu\text{C}$. [1]
- c) Two-point charges $+2\mu\text{C}$ and $+8\mu\text{C}$ are placed 10 cm apart in air. Sketch qualitatively the electric field lines, indicate the region and calculate the position where the resultant field will be zero. [3]
18. a) Write the conditions for validity of Coulomb's law of force between charges. [1]

- b) State and explain the principle of superposition of Coulomb's force with suitable diagram. [2]
- c) A polythene piece rubbed with wool is found to have a -ve charge of $0.64 \mu\text{C}$. (i) estimate the number of electrons transferred from which to which? (ii) is there a transfer of mass from wool to polythene? If yes, find the amount of mass transferred. [2]
19. a) Find the dimensions of Planck's constant. If its value in cgs system is $6.62 \times 10^{-27} \text{ erg-sec}$, what will be its value in MKS system? [2]
- b) Assuming that the mass m of the largest stone that can be moved by a flowing river depends upon velocity v , the density ρ and acceleration due to gravity g , show that mass m varies directly as the sixth power of the velocity of the flow. [3]

GROUP 'C' [3×8=24]

Long answer questions:

20. a) What is meant by resolution of a vector? Why resolving a vector is important in physics? Can the walking of a man be an example of resolution of vectors? Explain. [3]
- b) Can the magnitude of the rectangular component of a vector be greater than the magnitude of that vector. Explain. [1]
- c) A man rows a boat with a speed of 18 kmh^{-1} in the north-west direction. The shoreline makes an angle of 15° south of west. Find the components of velocity of the boat along the shoreline and perpendicular to the shoreline. [2]
- d) An athlete runs 2km, east and then 3km, 40° south of east. Find the magnitude and direction of total displacement. [2]
21. a) What do you mean by absolute zero and absolute scale of temperature? [2]
- b) A student claimed that the thermometers register their own temperature therefore, they are useless. How will you respond? [2]

FIRST TERM EXAMINATION 2082

Class: XI 'Science'

SET 'B'

F.M.: 50

Time: 2 hrs.

Subject: Computer Science (4271)

P.M.: 20

Candidates are required to give their answer in their own words as far as practicable.
The figure in the margin indicated full marks:

Attempt all the questions.

GROUP 'A' [9×1=9]

MULTIPLE CHOICE QUESTIONS.

1. The three main components of computer are
 - a) CPU, Memory, Tape
 - b) I/O, Printer, Mouse
 - b) Tape, I/O, Floppy disk
 - d) CPU, I/O, Memory
2. Which type of computer is smaller than a mainframe but larger than a microcomputer, and used by medium-sized organizations?
 - a) Hybrid Computers
 - b) Super Computers
 - c) Mini Computers
 - d) Analog Computers
3. Which is the major component of fourth generation computers?
 - a) Transistor
 - b) Microprocessor
 - c) Vacuum Tube
 - d) IC
4. A speaker is an example of device.
 - a) processing
 - b) storage
 - c) input
 - d) output
5. What is the binary equivalent of a decimal number 31?
 - a) 10101
 - b) 11111
 - c) 11010
 - d) 101101
6. How many bits are there in 2.5 kilobytes?
 - a) 20,480 bits
 - b) 25,600 bits
 - c) 40,960 bits
 - d) 20,480 bytes
7. The maximum number of bits used to represent hexadecimal number in binary is
 - a) 3
 - b) 4
 - c) 8
 - d) 16
8. The _____ is a small input device used to select and display objects on a screen.
 - a) Ink
 - b) Magnet
 - c) Light pen
 - d) None of these
9. Which of the following is not the font style?
 - a) Bold
 - b) Italics
 - c) Regular
 - d) Superscript

1 byte = 8 bits
1024 bytes

GROUP 'B' [5×5=25]

SHORT ANSWER TYPE QUESTIONS.

10. Why we use multimedia? Define linear and non-linear multimedia with examples. Explain the different components of multimedia. [1+1+3]
11. Explain the memory hierarchy in a computer system with a neat and labeled diagram. Describe the different levels of memory and discuss their characteristics in terms of speed, cost, and capacity. [2.5+2.5]
12. Describe any three types of positional number system. Convert the following as indicated. [3+1+1]

a) $(93.6)_{10} = (?)_2$

b) $(BAD.DAD)_{16} = (?)_8$

'OR'

Subtract 1011_2 from 10101_2 using 1's and 2's complement methods. [2.5+2.5]

13. What is mobile computing? Suppose the internet and mobile networks suddenly stopped working for one full week. Describe at least two problems people would face in education, business, and daily life. Suggest at least two alternative solutions people could use during that time. [1+1+1+1+1]

'OR'

Explain in detail the concept of mail merge in Microsoft Word. What are its primary benefits and disadvantages? [5]

14. Define output devices with examples. Write their functions. Write any four differences between CRT monitor and LCD monitor. [1+2+2]

GROUP 'C' [2×8=16]

LONG ANSWER TYPE QUESTIONS.

15. What do you mean by computer architecture? Describe each functional components of computer architecture with typical diagram. [1+7]
16. You are given to prepare a project report in MS Word about "Generations of Computers."
- a) Explain how you would use any four Word features (like Table of Contents, Page Numbering, WordArt, Styles, SmartArt, etc.) to organize the report neatly. [4×1.5=6]
- b) Write why these features are useful for readers. [2]

"Best of Luck"

GROUP 'B' [5×5=25]

SHORT ANSWER TYPE QUESTIONS.

10. Why we use multimedia? Define linear and non-linear multimedia with examples. Explain the different components of multimedia. [1+1+3]
11. Explain the memory hierarchy in a computer system with a neat and labeled diagram. Describe the different levels of memory and discuss their characteristics in terms of speed, cost, and capacity. [2.5+2.5]
12. Describe any three types of positional number system. Convert the following as indicated. [3+1+1]

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- a) Explain how you would use any four Word features (like Table of Contents, Page Numbering, WordArt, Styles, SmartArt, etc.) to organize the report neatly. [4×1.5=6]
- b) Write why these features are useful for readers. [2]

"Best of Luck"

KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Class: XI 'Science'

Time: 3 hrs.

Subject: English [0031]

F.M.: 75

P.M.: 30

Candidates are required to give their answer in their own words as far as practicable. The figure in the margin indicated full marks.

Attempt all the questions.

1. Read the following text and do the activities that follow.

Jay Tee, 30, has been a drug addict since he was a teenager. He has tried to kick the habit many times; going in and out of rehabilitation centers. He managed to stay clean for a month before suffering a **relapse**. There are many drug addicts like Jay Tee. Why are they into it in the first place? It doesn't matter if you are white or black or Asian or Latin, every race has their share of drug abuse problems.

Some people say that it is due to the **perception** of drugs being "bad and illegal" and therefore something so irresistible that you just have to try it. Somehow, teenagers are more curious to discover the "forbidden fruit". Some manage to kick the habit early but there are many others who are trapped.

For many teenagers, drug abuse is a mere escape. They use drugs to get out of their misery, pain, confusion and loneliness. Some teenage drug addicts grow up without guidance from their parents, and so they resort to drugs. Some are physically or mentally abused and drugs become the only thing that they are able to count on. Many succumb to drug addiction because of the need to feel like they belong. They try drugs to please their friends or to be part of a group. Many teenage drug abusers are also low achievers in schools and have low self-esteem.

There are also teenagers who claim that they were first exposed to different kinds of drugs at parties and **discos**. These drugs were distributed freely the first few times and they kept coming back for more. As a result, many of them became addicted even before they knew it. In certain cases, these teenagers might resort to committing crimes to support their expensive habit.

One effective tool in the fight against teen drug abuse is communication. Troubled teens need some form of outlet from the problems or depression they are facing. Parents should take time out from their busy schedules to spend time with their children and listen to their problems. Communication, however, should go beyond words. Parents should show their children that they really care. Adequate parental **supervision** has also

- d) Many teenagers use drugs as a way to escape pain, confusion, and loneliness.
 i. happiness ii. misery iii. success
- e) Teenagers may first be exposed to drugs at and discos.
 i. schools ii. parties iii. homes

2. Write short answers to the following questions in about 75 words each. [5×2=10]

- a) Explain the significance of the seasons in the story, *The Selfish Giant*.
 b) How is the poem, 'A Red, Red Rose' a romantic poem?
 OR
 Why is it necessary for the younger generation to take responsibility for carrying on traditions. Discuss. (*Sharing Traditions*)
 c) The poem, *Corona Says* reflects ecological awareness. Discuss.
 d) What does the line "*The Earth is not your property alone*" mean? (*Corona Says*)
 OR
 What is a hyperbole? Explain any two hyperboles used in the poem, "A Red, Red Rose."
 e) What did the Giant build around his garden? Why? (*The Selfish Giant*)

3. Write long answers to the following questions in about 150 words each. [2×5=10]

- a) The poem, *A Red, Red Rose* celebrates love as eternal and unchanging. Discuss.
 OR
 Why is it difficult to transmit the oral traditions in the modern world of globalization and technology? Describe. (*Sharing Tradition*)
 b) What moral lessons do we learn from "The Selfish Giant"? Explain.

4. Write a few paragraphs either in favour of or against the ban on social media. [7]

5. Write a letter to your cousin who is currently working in Korea encouraging him to return to Nepal and use the valuable skills and experiences he has gained abroad for the development of the nation. Also, remind him of the growing opportunities at home and the importance of contributing to the progress of the homeland instead of staying away for a long time. [8]

6. Traditionally, politics has been dominated by older leaders, believed to have the wisdom and experience to govern. Recently, many young people want to participate in politics, claiming they bring energy, innovation, and fresh ideas. This raises a debate: should young people take an active role in

politics, or should leadership remain with the experienced elderly. Write an essay discussing the advantages and disadvantages of allowing young people to involve in politics. Support your answer with examples and give your opinion. [10]

7. Do as indicated: [10x1=10]

- a) Talk to the children (polite/politely/politeness). (Choose the best option)
- b) I was **shocked** to hear the news. (Identify the word class of the bold letter word)
- c) Mina will pass the exam surely. (Rewrite the sentence putting the adverb in correct place)
- d) He often visits us. / Often he visits us. (Choose the best option)
- e) There is a book (onto/into/on) the desk. (Choose the best option)
- f) The cat jumped the bar and landed in the field. (Fill in the blank with a suitable preposition)
- g) She (seek) advice from her teacher before making the decision. (Rewrite the sentence in simple past tense)
- h) I think she (pay) the bill tomorrow. (Rewrite the sentence using correct tense)
- i) The monkey on the circus rope has lost the balance. It (fall) on the ground. (Rewrite the sentence using correct tense)
- j) If I (be) you, I'd visit the doctor. (Rewrite the sentence using correct tense)

8. Choose the best options: [5x1=5]

- a) The new rules were mandatory for all students. The meaning of the word mandatory is:
i. optional ii. compulsory iii. flexible iv. unnecessary
- b) Which word comes first, if gate, gain, game, garden, gas are arranged in alphabetical order?
i. gate ii. gain iii. gas iv. garden
- c) The abbreviation v. in a dictionary stands for:
i. verb ii. value iii. verify iv. various
- d) She looked at me when I told the truth.
i. angrily ii. angry iii. anger iv. angrier
- e) The prefix sub- in the word submarine means:
i. above ii. under iii. opposite iv. again

Culture and tradition are the true heartbeat of our identity. Let's celebrate them in an authentic way and commit to preserving them.

Happy Tihar!

...*Best of Luck*...

कक्षा : ११ (विज्ञान सङ्काय)

पूर्णाङ्क : ७५

समय : ३ घण्टा

विषय : नेपाली (००११)

उत्तीर्णाङ्क : ३०

विद्यार्थीहरूले सकेसम्म आफ्नै शब्दमा उत्तर दिनुपर्ने छ । दायाँ किनारामा दिइएको बङ्कले पूर्णाङ्क जनाउँदछ ।

सबै प्रश्नहरू अनिवार्य छन् ।

१. दिइएको वाक्यका रेखाङ्कित वर्णहरूको उच्चारण प्रयत्न र घोषत्व छुट्याई लेख्नुहोस् : (३)
जीवनको एउटा तर महत्त्वपूर्ण अंशलाई समेट्ने विशिष्ट घटनाको सिलसिलेवार प्रस्तुति कथा हो ।

२. दिइएको वाक्यलाई शुद्ध गरी पुनर्लेखन गर्नुहोस् : (३)
जीवन एउटा प्रवाहपील नदी हो । यो सामुन्द्रिक नाविकहरू को यात्रा जस्तै हो जसलाई हामी निरन्तर दोडाइरहेका हुँदा रहेछौ ।

३. दिइएको अनुच्छेदमा रेखाङ्कित पदहरूको पदवर्ग पहिचान गरी लेख्नुहोस् : (३)
घटनाप्रधान समाचार प्रायः दृश्य विवरणमा केन्द्रित हुन्छ । प्रक्रियापरक समाचारमा घटनाको प्रकृतिबनुसार त्यसलाई सरसरती वर्णन मात्र नगरी तथ्यको विस्तृत व्याख्या गरिन्छ ।

४. दिइएका प्रश्नहरूको उत्तर दिनुहोस् । (२+२=४)

क) दिइएको अनुच्छेदबाट दुई दुई ओटा तत्सम र आगन्तुक शब्द पहिचान गरी लेख्नुहोस् :
जिल्ला प्रशासन कार्यालय ललितपुरका अनुसार माध्यमिक तहका कक्षा ११ र १२ का विद्यार्थीले कलेज डेसमा मोटरसाइकल चलाएमा कारवाही गर्ने कानुनी व्यवस्था भएपछि सडकमा जयाभावी सवारी साधन चलाउने र दुर्घटना हुने प्रवृत्तिमा कमी आएको छ । अभिभावकले पनि यस कार्यको स्वागत गरेका छन् ।

ख) दिइएका शब्दहरूलाई शब्दकोशीय क्रममा मिलाएर लेख्नुहोस् :
ज्ञान, क्षेत्र, विशूल, कुशल

५. तलको अनुच्छेदबाट तीन ओटा पारिभाषिक शब्द पहिचान गरी वाक्यमा प्रयोग गर्नुहोस् : (३)
हिजोभन्दा आज हरेक सङ्क्रामक रोगको उपचार खोपद्वारा सम्भव हुँदै गएको छ । देशमा कुपोषणबाट मृत्यु हुने बालबालिकाहरूको तथ्याङ्क सार्वजनिक गरिएको अभिलेखबाट थाहा पाउन सकिन्छ ।

अथवा

दिइएको अनुच्छेदबाट एउटा अनुकरणात्मक शब्द, एउटा उखान, एउटा टुक्का पहिचान गरी तिनलाई वाक्यमा प्रयोग गर्नुहोस् :

पद्माखेरि आफूलाई नआएका कुरा गरिमा फटाफट हजुरआमालाई सोध्न जान्थी तर हजुरआमालाई 'काला अक्षर भैसी बराबर' थिए । जे भए पनि 'आगो ताप्लु मुढाको, कुरा सुन्नु बुढाको' भनेर पवन पनि हजुरआमाको कुराबाट हरेक दिन केही न केही सिक्किरहन्थ्यो । हजुरआमाको नाक राख्नुपर्छ भनेर उनीहरू टुप्पी कसेर खुस्खुरु पढ्ने गर्थे ।

६. अनुच्छेदबाट तीन तीन ओटा उपसर्ग र प्रत्यय लागेका शब्द पहिचान गरी तिनको निर्माण प्रक्रिया देखाउनुहोस् : (३)

प्रमुख राजनीतिक दलहरूले गैरजिम्मेवार बनेर मुलुकलाई अनिर्णयको बन्दी बनाउनु हुँदैन । दलीय कर्तव्य र राष्ट्रिय जिम्मेवारीलाई बुझेर सकेसम्म चाँडै नै विकास दिने काममा केन्द्रित हुनु आवश्यक छ ।

७. दिइएको अनुच्छेदका वाक्यहरूलाई पुलिङ्ग भए स्वीलिङ्ग र स्वीलिङ्ग भए पुलिङ्गमा परिवर्तन गरी पुनर्लेखन गर्नुहोस् : (४)
गायकले गीत गायो । आमाले गीत सुन्नुभयो । गीत सुनेर आमा घरमा आउनु भयो । आमा आउँदा भाइ खुसी भयो ।

८. तल दिइएको अनुच्छेदका वाक्यलाई निम्न आदरमा बदल्नुहोस् । (४)
उहाँले मलाई चिन्नुहुन्छ । उहाँलाई मैले पनि चिन्दछु । उहाँसँगको पुरानो सम्बन्ध छ । उहाँलाई मैले बोलाउँछु ।

९. दुवै प्रश्नहरूको उत्तर दिनुहोस् : (४+४= ८)

क) दिइएका अनुच्छेद पढी सोधिएका प्रश्नहरूको उत्तर दिनुहोस् :

जीवनको एउटा तर महत्त्वपूर्ण अंशलाई समेट्ने विशिष्ट घटनाको सिलसिलेवार प्रस्तुति कथा हो । यिनै घटनाको कार्यकारण शृङ्खलालाई कथानक भनिन्छ । कथाको आकार र स्वरूप तयार पार्ने कार्यमा कथानकले महत्त्वपूर्ण भूमिका निर्वाह गर्ने भएकाले यसलाई कथाको मुख्य तत्त्वका रूपमा लिने गरिन्छ । कथामा घटना अधि बढाउने वा कार्यव्यापार सम्पन्न गर्न पात्रको प्रयोग गरिएको हुन्छ । पात्र मानवीय र मानवेतर गरी दुवै किसिमका हुन्छन् । कथा भन्ने पात्रलाई समाख्याता भनिन्छ । समाख्याता कथाभित्रकै पात्र पनि हुन सक्छ र कथाबाहिरको पनि हुन सक्छ । कथाको पात्र आफैले कथाको समाख्यान गरेको छ भने त्यसलाई प्रथम पुरुष दृष्टिबिन्दु भएको कथा र अन्य पात्रका माध्यमबाट कथाको समाख्यान गरिएको छ भने त्यसलाई तृतीय पुरुष दृष्टिबिन्दुको प्रयोग भएको कथा भनिन्छ । प्रथम पुरुष दृष्टिबिन्दु भएका कथामा 'म', र 'हामी' जस्ता प्रथम पुरुषलाई बुझाउने पदको प्रयोग गरिएको हुन्छ भने तृतीय पुरुष दृष्टिबिन्दु भएका कथामा 'त्यो' ऊ, तिनीहरू आदि तेस्रो पुरुष बुझाउने पदको प्रयोग गरिएको हुन्छ । घटनाको समय, स्थान र त्यसबाट उत्पन्न हुने वातावरणलाई परिवेश भनिन्छ । कथाले भन्न खोजेको मूल कुरा सारवस्तु हो । यसैलाई सन्देश पनि भनिन्छ । यी सबैलाई प्रस्तुत गर्ने माध्यम चाहिँ भाषा हो । कथामा गद्य भाषाको प्रयोग हुन्छ । कथालाई प्रभावकारी रूपमा अधि बढाउन पात्रका बिचमा संवाद पनि हुन सक्छ । कथाको भाषालाई वर्णनात्मक, पत्रात्मक, संस्मरणात्मक, नाटकीय आदि शैलीमा प्रस्तुत गर्न सकिन्छ । यसैले कथानक, पात्र, परिवेश, दृष्टिबिन्दु, भाषाशैली र सारवस्तु यी सबै कथाका तत्व हुन् ।

प्रश्नहरू

(क) कथा भनेको के हो ?

(ख) कथामा कति किसिमका पात्र हुन्छन् ?

(ग) प्रथम पुरुष दृष्टिबिन्दु भएका कथामा कस्ता पदको प्रयोग गरिएको हुन्छ ?

(घ) कथाका तत्व के के हुन् ?

(ख) दिइएको अनुच्छेद पढी सोधिएका प्रश्नको उत्तर लेख्नुहोस् :

रातको बाह्र बजेको थियो । कोठामा म थिएँ । स्वास्नी थिई विरामी र सोचिरहेका थिएँ, अप्रेसनका लागि जुटाउनुपर्ने पाँच लाख रुपियाँ कहाँबाट ल्याउने ? एकाएक प्रकट भए भगवान् कोठामा । भस्किर्रँ । के भयो ! के यो सत्य त हो ! चिमोटे आफैँलाई । रहेनछ यो कुनै सपना वा भ्रान्ति । मुस्कुराउँदै भने भगवान्ले- 'ए मनुवा ! के माग्छस् माग् ।' यो अप्रत्यासित क्षण । म के मागूँ ?

छिटो माग ।' भने भगवान्ले । हड्बडिएँ सायद । के माग्ने ? बिलिडड, गाडी, पैसा । चाहिएको छ अप्रेसनका लागि खुसी जीवनका लागि । तर कति माग्ने ? दश लाख, पचास लाख, एक करोड, दश करोड ? 'छिटो माग । समय सकिन लाग्यो ।' भगवान्ले घडी हेर्दै भने । 'एकै सासमा भने- दश करोड सपियाँ दिनुस् प्रभु !' 'तथास्तु !' भगवान् विलाए । एकाएक थुप्रिएँ करोडका नोटहरू । खुसीले उफ्रिएँ र उठाउन खोजेँ स्वास्नी चिसो भइसकेकी थिई । नाडी छामे । थिएन नाडी । थियो स्वास्नीको लास एकातिर । अर्कोतिर नोटका बिटाहरू थिए अनि थिए म बिचमा अवाक् । चिच्याएँ- प्रभु ! मलाई स्वास्नीको जीवन दिनुस् । तर थिएनन् भगवान् त्यहाँ । थियो शून्यता । केवल शून्यता । थिए नोटका बिटाहरू ।

प्रश्नहरू :

क) प्रस्तुत कथामा वर्णित घटना कुन समयमा घटेको हो ?

अ) रातको एघार बजे

आ) रातको बाइ बजे

इ) रातको एक बजे

ई) रातको साढे बाइ बजे

ख) अप्रेसन खर्च कति लाग्ने रहेछ ?

अ) पाँच लाख

आ) एक करोड

इ) दश करोड

ई) दश लाख

ग) एकाएक प्रकट भएका भगवान्ले के भने ?

अ) ए ! मनुवा के सोच्दै छस् ?

आ) ए मनुवा ! तलाई के चाहियो ?

इ) ए मनुवा ! के माग्छस् माग् ।

ई) ए भाइ, म के मागूँ ?

घ) भगवान् के भनेर विलाए ?

अ) तथास्तु

आ) धन्यवाद

इ) तिम्रो इच्छा पूरा भयो

ई) नमस्कार

१०. अनुच्छेदबाट चार ओटा बुँदा टिपी एक तृतीयांशमा सारांश लेख्नुहोस् ।

(२+२=४)

नेपाल खुला एवं बजारोन्मुख अर्थव्यवस्था भएको मुलुक हो । हाम्रो देशले आफ्नै स्रोतसाधनको पहिचान गरी आर्थिक विकास र गरिवी निवारणको वाणिज्य नीति अवलम्बन गरेको छ । अन्तर्राष्ट्रियस्तरमा प्रतिस्पर्धा गर्नु र व्यापार व्यवसायको प्रवर्धन गर्नु यसको उद्देश्य हो । सामान निकासी, पैठारीलाई समयानुकूल बनाई राजस्व सङ्कलन गर्दा देशको अर्थतन्त्रमा टेवा पुग्छ । आफ्नो देशमा उत्पादित जडिवुटी तथा कृषिजन्य वस्तुलाई व्यवस्थापन गरी आर्जन गर्न सकिने ढाँचामा हाम्रो वाणिज्य नीति तय भएको छ । व्यापार घाटा कम गर्न र रोजगारीको संख्यामा वृद्धि गर्न यस नीतिले सहयोग पुऱ्याउने अपेक्षा लिइएको छ । प्राकृतिक र जैविक विविधताले युक्त यस देशमा पशुपन्छी र वनस्पतिजन्य उत्पादनलाई उचित प्रविधि र जनशक्तिको संयोजनद्वारा निर्यात प्रवर्धन गर्न सकिने सम्भावना टरेको छैन । लघु उद्योग, घरेलु उद्योगलाई प्रोत्साहन गर्दै नेपाली माटो सुहाउँदो अर्थतन्त्र निर्माणतिर लाग्ने हो भने राज्यको राजस्व संकलनमा पनि वृद्धि हुन्छ । देशभित्रै रोजगारी सिर्जना गर्नका लागि उद्योग व्यवसाय सञ्चालनार्थ राज्यले निजी क्षेत्रलाई पनि प्रोत्साहन गर्न सके हाम्रो अर्थव्यवस्था मजबुत बन्न सक्छ । यसरी सरकारले सहजकर्ताको भूमिका निर्वाह गरी उत्प्रेरक र नियम निकायबाट अनुगमनसमेत गर्दै निजी क्षेत्रको पनि सक्रिय सहभागितामा देशलाई आर्थिक समृद्धिको बाटोमा हिँडाउन सकिन्छ ।

११. पुस्तकालयमा आवश्यक पुस्तकको माग गर्दै विद्यालयका प्रधान अध्यापकलाई एउटा निवेदन लेख्नुहोस् ।

(४)

१२. नेपालमा बढ्दो विमान दुर्घटना शीर्षकमा एक अनुच्छेद लेख्नुहोस् ।

(४)

१३. छाउपडी प्रथा अन्त्य गर्न के गर्नु पर्ला ? आफ्नो प्रतिक्रिया लेख्नुहोस् ।

(४)

१४. कुनै दुई प्रश्नको उत्तर लेख्नुहोस् ।

(४+४=८)

क) विइएको कवितांश पढी सोधिएका प्रश्नहरूको विनुहोस् :

डिगर्चामा डोब तिम्रो चिसो हिउँभित्र होला
वेवावती किनारभरि पौरखको चिनो होला
वीर पुर्खा ! तिम्रीलाई मितेरीले मात्रै बाँध्यो
सागर तरी संसारभरि वीर गोर्खा रगत बग्यो ।

प्रश्नहरू :

(अ) माथिको कवितांशमा हाम्रा पुर्खाहरूका बारेमा के भनिएको छ ?

(आ) हामीले हाम्रा पुर्खाको गौरव कसरी जोगाउन सक्छौं ?

ख) तलको कथांश पढी सोधिएका प्रश्नको उत्तर विनुहोस् :

खोलो अचेल फेरि गाउँतिर सोभिएको रहेछ । यो खोलाले पनि गाउँलाई धुक्कै रुवायो । हुन त रुवायो मात्रै भन्नु खोलाप्रति अन्याय ठहर्ला । यो खोलाले गाउँलाई पालेको पनि छ । हाम्रो विगतको लगभग आधा उमेर यही खोलाको ढिल र बगरहरूमा बितेको थियो । कहिले ओरहा खाने निहुँमा त कहिले गाईभैँसी चराउने निहुँमा । खोलामा बाढी आएको बेला हामी गाईको पुच्छर समाएर बाढी तर्ने गर्थ्यौं अथवा यस्तो बेलामा हामी खोलाको किनारमा बसेर बाढीको आनन्द लुट्थ्यौं । तर्न डराउने कोही मानिस आए भने हामी तीनचार जना एकअर्काको हात समाउँदै बाढीमा हेलिन्थ्यौं र किनारमा पस्निरहेको मानिसलाई तारिदिन्थ्यौं र यस्तो काममा खास गरी परिछनकै विशेष रुचि थियो । “यहीनिर कुवा छ, गए सालको वर्षामा मूल फुटेको एकदम चिसो र मिठो पानी छ ।” ऊ पानी लिएर आइसकेको थियो ।

प्रश्नहरू :

(अ) खोलाले रुवायो भन्ने कुरा मात्र गर्दा कसरी अन्याय हुन्छ ?

(आ) ‘म’ पात्रका खोलासँगका बालअनुभूति कस्ता छन् ?

ग) विइएको निबन्धांश पढी सोधिएका प्रश्नहरूको छोटो उत्तर लेख्नुहोस् :

संस्कृति हाम्रो परिचय हो, हाम्रो पहिचान हो र हाम्रो अनुभूति हो । संस्कृति हाम्रो बोल्ने, विचार गर्ने र बाँच्ने तरिका पनि हो । हाम्रो संस्कृति बाँच्नुपर्छ, त्यसैले हाम्रा सांस्कृतिक मूल्यहरू बाँच्नुपर्छ । दसैँ हाम्रो संस्कृति हो, तिहार हाम्रो संस्कृति हो । तिज, बुद्ध पूर्णिमा, उधौली, उमौली, ल्होसार, क्रिसमस, इद सबै हाम्रा संस्कृति हुन् । सम्यताको यस यात्रामा म बारम्बार सोच्छु र सोच्छु यस्ता सुन्दर अवसरलाई हामीले सदुपयोग गर्ने पर्छ । संस्कृतिका नाममा लिङ्ग, वर्ग, जातिका आधारमा हुने भेद विकृति हुन् । ती कुरा हट्नुपर्छ र संविधानले यसको सुनिश्चितता पनि गरेको छ ।

प्रश्नहरू :

(अ) संस्कृति भनेको के हो ?

(आ) संविधानले सुनिश्चित गरेका सांस्कृतिक विषयहरू के के हुन् ?

१५. कुनै एक प्रश्नको समीक्षात्मक उत्तर दिनुहोस् ।

(८)

(क) ‘गाउँको माया’ कथामा वर्गीय द्वन्द्वको चित्रण कसरी गरिएको छ ? समीक्षात्मक उत्तर लेख्नुहोस् ।

(ख) वीर पुर्खाहरूले कसरी नेपालको रचना गरे ? समीक्षात्मक उत्तर लेख्नुहोस् ।

१६. विइएका मध्ये कुनै एक शीर्षकमा २५० शब्दसम्ममा नघटाई निबन्ध लेख्नुहोस् ।

(८)

(क) वातावरण संरक्षणमा मेरो भूमिका

(ख) मलाई मन पर्ने साहित्यकार

(ग) नेपालमा जलस्रोत

सफलताको शुभकामना !!!

FIRST TERM EXAMINATION 2082

Class: XII 'Science'

SET-B

Time: 25 min

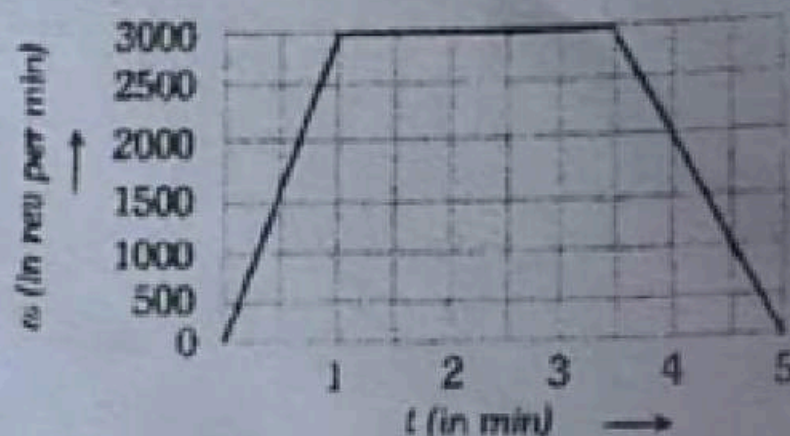
Subject: Physics [0031]

Re-write the best option among the four options on your answer sheet.

Attempt all the questions.

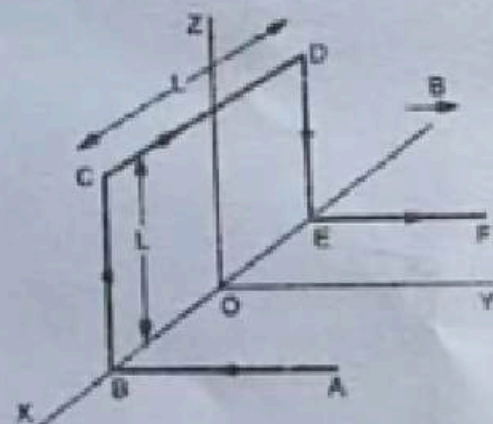
GROUP 'A' MCQ [11×1=11]

1. As a part of maintenance inspection, the compressor of a jet engine is made to spin according to the graph as shown. The number of revolutions made by the compressor during the test is:



- a) 11250 b) 12750 c) 16570 d) 9000
2. Two particles execute SHM of same frequency and amplitude along the same straight line. They pass one another while going in opposite directions. Each time their ~~amplitude~~ ^{displacement} is one-third of their ~~amplitude~~ ^{amplitude}. The phase difference between them is: ~~displacement~~
- a) 141° b) 35.25° c) 70.5° d) 105.75°
3. A torque τ is applied to a uniform disc of radius R and it rotates through an angle θ . The work done by the torque is::
- a) $\tau R^2 \theta$ b) $\tau R \theta$ c) $\tau \theta$ d) $\frac{\tau \theta}{2}$
4. A mass-spring system oscillates with a certain period on Earth. If the spring constant is doubled, the time period of oscillation will:
- a) Double b) Decrease by a factor of $\sqrt{2}$ c) Increase by a factor of $\sqrt{2}$ d) Halve
5. The wavelength the of a sound wave is λ and its velocity is v . If wavelength is increased to 4λ , the velcoity of the wave will be
- a) v b) $2v$ c) $4v$ d) $v/4$

6. In a hydrogen-like atom, the radius of the n^{th} orbit is r_n and the speed of the electron in that orbit is v_n . If T_n and V_n denote the kinetic and potential energies of the electron in the n^{th} orbit, then:
- a) $T_n \propto r_n$ ☒ b) $T_n \propto \frac{1}{r_n}$ c) $V_n \propto r_n$ d) $V_n \propto \frac{1}{r_n^2}$
7. Which of the following property of wave is independent of the other?
- ☒ a) amplitude b) wavelength
c) velocity d) frequency wavelength
8. When a long steel pipe is tapped at one end, a listener hears two sounds:
- a) of the same intensity, at same time
b) of the same intensity, one after another
c) less intense first, more intense later
☒ d) more intense first and less intense later
9. An air column closed at one end is set into vibration and emits a fundamental note of 550 cycles per second. If the speed of sound in air is 330 m/s, the wavelength of the sound in the surrounding air is:
- a) 30 cm ☒ b) 60 cm c) 15 cm d) 45 cm
10. A wire ABCDEF is bent in the form as shown in the figure. The wire carries a current I and is placed in a uniform magnetic field of induction B parallel to positive Y-axis. If each side is of length L , the force experienced by the wire will be:
- a) $2IBL$ along the positive Z-direction
b) $2IBL$ along the negative Z-direction
c) IBL along the positive Z-direction
☒ d) IBL along the negative Z-direction
11. An proton is projected in the direction parallel to both electric and magnetic field, proton will
- a) magnitude of velocity goes on decreasing
☒ b) magnitude of velocity goes on increasing
c) There will be no change in magnitude of velocity.
d) At first magnitude of velocity decreases and then increases



Class: XII 'Science'

F.M.: 75

Time: 3 hrs.

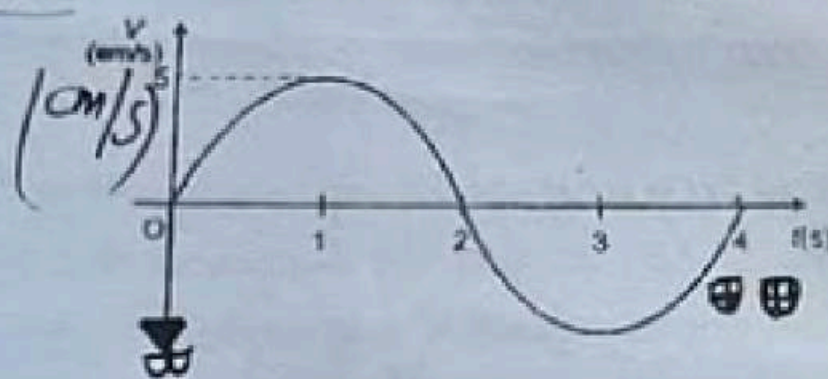
Subject: Physics [1021]

P.M.: 30

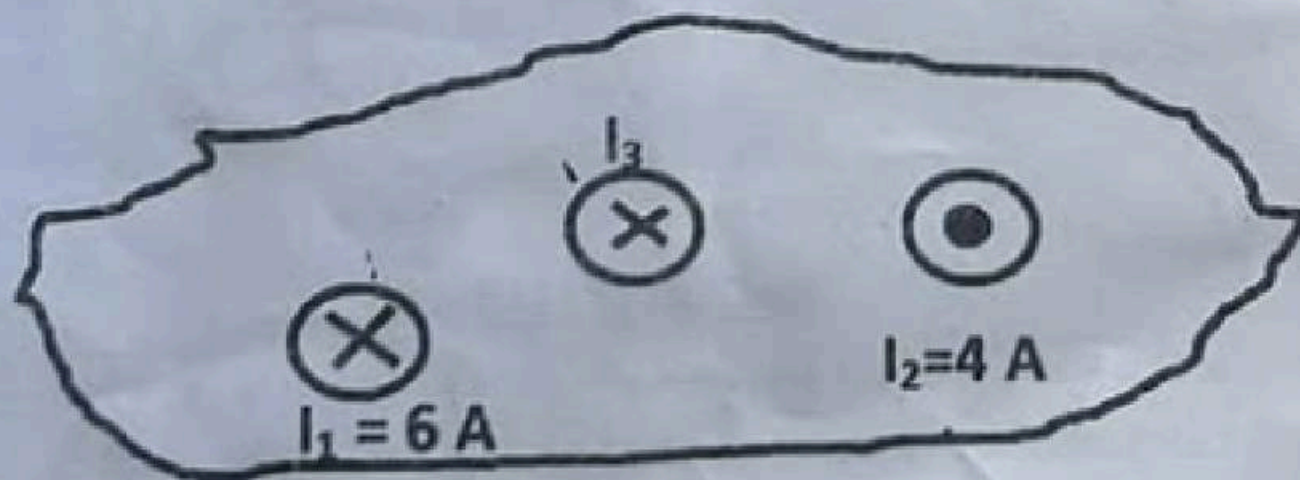
*Candidates are required to give their answer in their own words as far as practicable.**The figure in the margin indicated full marks:***Attempt all the questions.****GROUP 'B' [8×5=40]****Short answer questions:**

12. a) Does the moment of inertia of a body become zero as soon as it stops rotating? Explain. [1]
b) Is the radius of gyration of a body a constant quantity? If not then, on which factors does it depend? [1]
c) Derive an expression for moment of inertia of a thin uniform rod about an axis passing through one end and perpendicular to its length. What will be the result if the axis passes through the center? [3]
13. a) State and prove the principle of conservation of angular momentum with suitable example. [2]
b) Why the middle seat is more comfortable while travelling in a bus in a bumpy road? [1]
c) A flywheel of moment of inertia 0.32 Kg m^2 is rotated steadily at 120 rad s^{-1} by a 50 W electric motor (i) Find the kinetic energy and angular momentum of the flywheel. (ii) Calculate the value of the frictional couple opposing the rotation. [2]
14. a) Write the characteristics of simple harmonic motion. [1]
b) Show mathematically and graphically the total energy of a particle executing SHM is constant. [3]
c) A particle executes SHM between $x = \pm 0.1 \text{ m}$. When its displacement is $x = 0.05 \text{ m}$, its speed is 0.6 m/s . Find the total energy if $m = 0.2 \text{ kg}$. [1]
15. a) What is the principle of superposition of waves? Discuss the result of superposing of two waves of equal amplitude and same frequency travelling in opposite direction. [3]

- b) A certain transverse sinusoidal wave of wavelength 20 cm is moving in the positive x direction. The transverse velocity of the particle at $x = 0$ as a function of time is shown. Find
- i) the amplitude of the wave and ii) write the displacement equation that describes the wave. [2]



16. a) Write the expression for the magnitude of the magnetic Lorentz force, stating the physical quantities involved. Also, explain the Fleming's left hand rule used to determine the direction of the force. [2]
- b) A proton moves in a uniform magnetic field of strength 0.500 T directed along the x-axis. At initial time, $t = 0$ s, the proton has velocity, $\vec{v} = (1.95 \times 10^5 \hat{i} + 2.00 \times 10^5 \hat{k})$ m/s. Find
- (i) At initial time, what is the acceleration of the proton?
- (ii) Is the path circular or helical? If helical, calculate the radius of helical trajectory and also calculate the pitch of the helix. [3]
17. a) What is toroid? Use ~~Ampere's~~ Ampere's circuital law to find out the magnetic field inside a toroid carrying current. [3]
- b) The value of line integral of $\vec{B}$ around the closed path as shown in figure is 3.77×10^{-6} Tm. Find the value of I_3 ? [2]

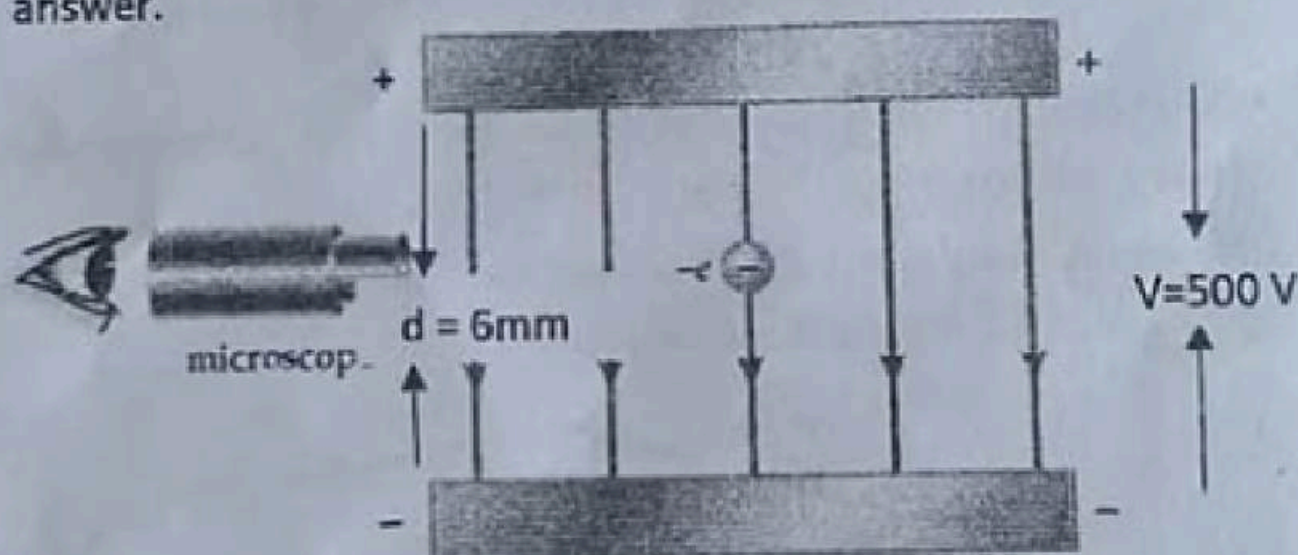


18. a) State the postulates of Bohr's theory of hydrogen atom. [2]

- b) Show that the speed of an electron in the first excited atom of hydrogen atom is $1/274$ times the speed of light. [2]
- c) Explain why the electron moves relatively faster in the inner orbit as compared to the outer orbit? [1]

19. Millikan's oil drop experiment was replicated in the laboratory. The droplet radius was 7×10^{-7} m, and the density of the oil was 900 kg m^{-3} . When a potential difference of 500 V was applied to the plates, the oil droplet was observed to be stationary.

- a) Calculate the charge on the oil drop. [2]
- b) Comment on the most likely measurement to be responsible for the discrepancy between the calculated charge and the accepted value of the charge on the electron. [1]
- c) If the potential difference is raised above 500 V what will happen to the oil drop? [1]
- d) If the polarity of the plates is reversed while keeping the potential difference at 500 V, what will happen to the oil drop? Explain your answer. [1]



GROUP 'C' [3×8=24]

20. During a thunderstorm, a student observes lightning first and hears thunder after 3 seconds. The temperature of air is 25°C . The speed of sound in air at 0°C is 332 m/s .
- a) Derive the expression for the speed of sound in air starting from the assumptions of Laplace's correction in terms of temperature. [3]
- b) Calculate the distance of the lightning strike from the observer. [2]

- c) Discuss how humidity affects the speed of sound in air. [1]
- d) "What do you mean by harmonics? Is the statement 'All harmonics are overtones, but all overtones are not harmonics' correct? If not, explain." [1]



21. a) A current element produces a magnetic field even though it is electrically neutral. Explain how this is possible. [1]
- b) State Biot-Savart law. [1]
- c) Use Biot-Savart law to find out the magnetic field at a point due to a long straight conductor carrying current. [3]
- b) A closely wound, circular coil with radius 2.40 cm has 800 turns. (i) What must be the current in the coil be if the magnetic field at the center of the coil is 0.0580 T? ii) At what distance from the center of the coil on the axes of the coil, the magnetic field is $1/8$ times that at the center of the coil? [3]

22. a) The work function of a metal is 6.4×10^{-19} . Explain what is meant by the term work function. Light with frequency of 1.2×10^{15} Hz is shone onto the metal surface. Find whether or not the photons of this light will cause photoelectric effect to take place. [2]
- c) Draw a graph showing the variation of stopping potential with frequency of radiation incident on a metal plate. How can the value of Planck's constant be determined from the graph? [3]
- d) From the following data, plot a graph for the photo-electric emission and find the value of threshold frequency, work function and Planck's constant? [3]

Frequency of incident light, f	3×10^{14} Hz	4×10^{14} Hz	5×10^{14} Hz	6×10^{14} Hz	7×10^{14} Hz
Maximum K.E of emitted photo electrons, E	0.1 eV	0.5 eV	0.9 eV	1.3 eV	1.7 eV



FIRST TERM EXAMINATION 2082

Class: XII 'Science'

Time: 25 Min.

SET- 'A'

Subject: Chemistry (3021)

Tick the best options among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

GROUP - 'A' (MCQ) [11×1 = 11]

1. Volume of water be added to $N/2$ HCl to prepare 500 cm³ of $N/10$ solution is
a) 100 ml b) 200 ml c) 300 ml d) 400 ml
2. Which of the following solutions of sulphuric acid will exactly neutralize 25 ml of 0.2 M sodium hydroxide solution?
a) 12.5 ml of a 0.1 M solution b) 25 ml of a 0.1 M solution
c) 25 ml of a 0.2 M solution d) 50 ml of a 0.2 M solution
3. Equivalent weight of Mohr's salt if it acts as reducing agent is
a) Mol.wt. b) Mol.wt./3 c) Mol.wt./5 d) Mol.wt./6
4. Which of the following salts does not undergo hydrolysis?
a) $CuSO_4$ b) $FeCl_3$ c) NH_4Cl d) Na_2SO_4
5. Solubility of $AgCl$ will be minimum in
a) 0.01 M $NaCl$ b) 0.01 M $CaCl_2$
c) pure water d) 0.001 M $AgNO_3$
6. Extremely hot copper wire reacts with steam to give
a) CuO b) Cu_2O c) Cu_2O_2 d) CuO_2
7. Iron is extracted on large scale from Fe_2O_3 by
a) Reduction with Al b) Smelting with carbon
c) Calcination d) Roasting
8. The purest form of iron is
a) White cast iron b) Grey cast iron
c) Wrought iron d) Steel

9. When chloroform is boiled with aqueous *KOH* followed by acidification, it gives

- a) Methoxy methane
- c) Methanoic acid

- b) Methanol
- d) Ethanol

10. The best reagent to convert alcohol into alkyl chloride is

- a) PCl_3
- c) $HCl/ZnCl_2$

- b) PCl_5
- d) $SOCl_2$

11. Chlorine gas reacts with benzaldehyde to give

- a) Chlorobenzene
- c) Benzal chloride

- b) Benzyl Chloride
- d) Benzoyl chloride

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

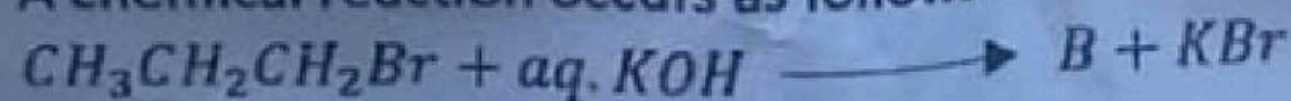
Attempt all the questions.

GROUP - 'B' [8×5=40]

Short answer questions

1. (a) Deduce normality equation. [2]
 b) Calculate the equivalent weight of $KMnO_4$ in acidic medium including reaction involved. [2]
 c) How would you apply titration in pharmaceutical industry? [1]
2. (a) Distinguish between titrant and titrand. [2]
 b) How many ml of a 0.1 M HCl are required to react completely with 1 gm mixture of Na_2CO_3 and $NaHCO_3$ containing equimolar amount of the two. [3]
3. (a) Differentiate between solubility product and ionic product. [2]
 b) If equal volume of two solutions having $pH = 1$ and $pH = 12$ are mixed together, what will be the pH of the resulting solution? [3]
4. A coinage metal (M) of electronic configuration $[Ar]4s^13d^{10}$ belongs to group IB in the periodic table.
 a) Draw a blast furnace for the smelting process during the extraction of (M) using its chief ore. [2]
 b) Explain the different chemical reactions involved during the formation of matte in the furnace. [3]
5. (a) What happens when nitroethane is reduced by
 i) H_2/Li ii) $LiAlH_4$ iii) Zn/NH_4Cl [1+1+1]
 b) Among *alc.* KNO_2 and *alc.* $AgNO_2$, which reagent is suitable for the preparation of nitroethane from chloroethane? Explain with reaction. [2]

A chemical reaction occurs as follows:



(A)

- a) Identify the compound 'B' in the above reaction and write its IUPAC name. [1]
- b) Which substitution reaction mechanism occurs in the above reaction and why? [1]

- c) What is the ...
 aq. KOH in the above reaction?
7. d) How can you obtain nitropropane from compound 'A'?
- a) Starting from trichloromethane, how would you prepare
 i) Methane ii) Chloropicrin iii) Chlorethane
- b) Why is chloroform stored in a dark brown airtight bottle?
- c) Why does chloroform not give white precipitate with aqueous silver nitrate?
8. a) Give an example of Williamson's ether synthesis. (1)
- b) Convert ethoxyethane into methoxyethane. ? ?
- c) What happens when ethoxyethane is heated with:
 i) excess HI ii) conc. H_2SO_4
- d) Why are old sample of ether not distilled to dryness in air?

GROUP - 'C' [3X8=24]

Long answer questions.

9. a) Derive expression of Ostwald's dilution law for weak electrolyte and state the law. (2)
- b) NH_3 acts as Bronsted base but not Arrhenius base, why? (1)
- c) Why is aqueous solution of $CuSO_4$ acidic in nature? (1)
- d) Define indicator. How will you select suitable indicator for the titration between weak acid and strong base? (1+2)
10. a) What is rusting? Describe the electrochemical theory of rusting of iron. (1+4)
- b) Write the application of common ion effect and solubility product principle in qualitative salt analysis. (3)
11. a) Write the chemical equations to illustrate
 i) Wurtz-Fittig reaction ii) Sandmeyer's reaction (1+1)
- b) Why are haloalkanes more reactive than haloarenes towards nucleophilic substitution reaction? (2)
- c) What happens when chlorobenzene is
 i) treated with chloral in acidic medium?
 ii) heated with $Ni - Al$ in alkaline medium? (2)
- d) Give proper reaction sequence to convert primary haloalkane into secondary haloalkane for the organic compound having molecular



Knowledge through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL

FIRST TERM EXAMINATION 2082

Class: XII 'Science'

SET-A

Time: 25 mins.

Subject: Com. Mathematics (0081)

Write the best option among the four options on your answer sheet

GROUP-A (MCQ) [11×1=11]

Re-write the correct option of each question in your answer sheet.

1. If $\frac{(n+2)!}{(n-1)!} = 720$, what is the value of n ?

A) 8 B) 9 C) 10 D) 11

What is the value of $\frac{(5000)!}{(2500)!}$?

A) $\frac{2^{2500}[1 \times 3 \times 5 \times \dots \times 5000]}{(2500)!}$

B) $2^{2500}[1 \times 3 \times 5 \times \dots \times 4999]$

C) $\frac{2^{2500}[1 \times 3 \times 5 \times \dots \times 4999]}{(2500)!}$

D) $2^{2500}[1 \times 3 \times 5 \times \dots \times 499]$

3. If $z = \sqrt{3} - 1$, then what is the $\arg(\bar{z})$?

A) 90° B) 180° C) 270° D) 0°

4. Which of the following systems of linear equations is *consistent* and *independent*?

A) $2x - 3y = 5; 4x - 6y = 10$

B) $2x - 3y = 5; 4x - 6y = 5$

C) $2x - 3y = 5; x + 6y = 5$

D) $x - 2y = 5; 3x - 6y = 10$

5. What is the surface area of a sphere if its volume is increasing at the same rate as its radius?

A) 1 unit

B) $\frac{1}{2\sqrt{\pi}}$ units

C) 4π units

D) $\frac{4\pi}{3}$ units

6. For which value of y is tangent to the curve $y = x + \frac{4}{x^2}$ is parallel to the x -axis?

A) $y = 2$

B) $y = 3$

C) $y = 0$

D) $y = 1$

7. What is the value of p , if $\lim_{x \rightarrow 0} \frac{e^{px} - \tan x - 1}{x} = 2$?
 A) 0 B) 1 C) 3 D) Does not exist

8. Which of the following represents the augmented matrix of the system of linear equations: $3x + 2y - 1 = 0$, $3y + 5x - 1 = 0$, $6y - 2x + 5 = 0$?

A) $\left(\begin{array}{cc|c} 3 & 0 & 2 \\ 5 & 3 & 0 \\ -2 & 6 & 0 \end{array} \right) \begin{array}{c} 1 \\ 1 \\ 5 \end{array}$

B) $\left(\begin{array}{cc|c} 3 & 0 & 2 \\ 5 & 3 & 0 \\ -2 & 6 & 0 \end{array} \right) \begin{array}{c} -1 \\ -1 \\ 5 \end{array}$

C) $\left(\begin{array}{cc|c} 3 & 0 & 2 \\ 5 & 3 & 0 \\ 6 & -2 & 0 \end{array} \right) \begin{array}{c} 1 \\ 1 \\ 5 \end{array}$

D) $\left(\begin{array}{cc|c} 3 & 0 & 2 \\ 5 & 3 & 0 \\ -2 & 6 & 0 \end{array} \right) \begin{array}{c} 1 \\ 1 \\ -5 \end{array}$

9. What is the value of integral $\int \frac{1}{\sqrt{3-2x^2}} dx$?

A) $\frac{-1}{\sqrt{2}} \cos^{-1} \frac{\sqrt{2}x}{3} + c$

B) $\frac{1}{\sqrt{2}} \sin^{-1} \sqrt{\frac{2}{3}} x + c$

C) $-\frac{1}{\sqrt{2}} \sin^{-1} \frac{\sqrt{2}}{3} x + c$

D) $-\frac{1}{\sqrt{2}} \sin^{-1} \sqrt{\frac{2}{3}} x + c$

10. For what is the value of p , does

$$\int \sqrt{p^2 - x^2} dx = \frac{x}{2} \sqrt{p^2 - x^2} + 2 \sin^2 \left(\frac{x}{p} \right) + C ?$$

A) 1

B) 2

C) 3

D) None of these

11. Which of the following represent the order and degree of the differential equation

$$\sqrt{\frac{d^2y}{dx^2}} = \sqrt[5]{1 + \frac{dy}{dx}} ?$$

A) 2, 5

B) 5, 2

C) 2, 2

D) 2, 10



KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Class: XII 'Science'
Time: 3hr

Subjects: Mathematics (0081)
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.
Attempt all the questions.

P.M. 75
P.M. 30

GROUP - 'B' [0x5=40]

Short Answer Questions.

12. a) What is the total number of distinct permutations of a set of n objects, if r of them are of the same kind and the rest are distinct?
b) What is the relationship between $P(n, n)$ and $P(n, n-1)$?
c) State addition principle of counting.
d) What is the exponential form of complex number $r(\cos \theta^\circ + i \sin \theta^\circ)$?
e) Write the necessary condition for having solution of the system of linear equations by Cramer's Rule.
[1+1+1+1+1=5]
13. a) How many words can be formed using all the letters of the word **EQUATION** so that
i) all the vowels are together?
ii) consonants occupy odd places?
b) How many even numbers of 3 different digits can be formed from the digits 0, 2, 3, 5, 6, 8?
[1+1]
14. Solve the following system of linear equations by Gauss elimination method.
 $2x + y + 4z - 12 = 0, 8x - 3y + 2z - 20 = 0, 4x + 11y - z - 33 = 0.$
[3]
15. a) Five boys and five girls sit alternately around round table. In how many ways can this be done?
[5]
[2]

b) Using Determinant method, solve the following system of linear equations.

$$2x + y - z = 9$$

$$3x - y + 2z = -1$$

$$4x + y - 3z = 17$$

16. a) Write the integral of $\int \operatorname{cosech} x \, dx$. [3]

b) If $\lim_{x \rightarrow 1} f(x) = 0$, $\lim_{x \rightarrow 1} g(x) = 0$ and $g'(1) \neq 0$, what is the relation between $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)}$ and $\lim_{x \rightarrow 1} \frac{f'(x)}{g'(x)}$? [1]

c) What is the relationship between derivative and antiderivative of function? [1]

d) What is the equation of tangent to the curve $y = g(x)$ at the point (h, k) that is parallel to Y -axis? [1]

e) What is the derivative of $\tanh^{-1} u$ with respect to u ? [1]

17. a) If n is natural number and multiple of 3, then prove that [1]

$$1 + \left(\frac{-1+i\sqrt{3}}{2}\right)^n + \left(\frac{-1+i\sqrt{3}}{2}\right)^{2n} = 3$$

[2]

b) Water flows into an inverted conical tank at the rate of $36 \text{ cm}^3/\text{min}$. When the depth of water is 12 cm, how fast is the level rising if the radius of base and height of the tank is 21 cm and 35 cm respectively.

[3]

18. a) If radius of circle is increasing from 7 cm to 7.1 cm, determine the approximate increase in area.

b) Evaluate: $\int \frac{t^2}{t^4 - 2t^2 - 15} dt$. [2]

[3]

19. a) Find $\frac{dy}{dx}$ if $e^{\sinh^{-1} x} = y$

[2]

b) If $\sqrt{1-x^2} dy + \sqrt{1-y^2} dx = 0$, show that $x\sqrt{1-y^2} + y\sqrt{1-x^2} = C$ where C is constant.

[3]

GROUP - 'C' [3x8=24]

20. a) Evaluate: $\frac{\left(\sin\frac{\pi}{8} - i \cos\frac{\pi}{8}\right)^9}{\left(\sin\frac{\pi}{8} + i \cos\frac{\pi}{8}\right)^9}$. [2]
- b) Use De-moivre's theorem to find cube roots of $8i$. [3]
- c) Are linear permutations and circular permutations interrelated? [3]
Justify your answer with a contextual example.
21. a) Find the points on the curve $\frac{x^2}{9} + \frac{y^2}{16} = 1$ at which the tangents are parallel to x - axis. [2]
- b) The rate of change of volume of sphere is proportional to square of its diameter. Explain [3]
- c) Evaluate: $\int \frac{1}{\cos x - \sqrt{3} \sin x} dx$. [3]
22. a) Solve: $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$. [2]
- b) State the necessary conditions under which the method of partial fraction decomposition is applicable for evaluating an integral. Justify your answer with suitable example. [3]
- c) Find the angle between the curves $y = x^2$ and $x = y^2$ at the point $(1,1)$. [3]

Kathmandu Model Secondary School
KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Class: XII 'Science'

SET 'A'

F.M.: 50

Time: 2 hrs.

Subject: Computer Science [4271]

P.M.: 20

Candidates are required to give their answer in their own words as far as practicable. The figure in the margin indicated full marks:

Attempt all the questions.

GROUP 'A' [9×1=9]

MULTIPLE CHOICE QUESTIONS.

1. What will be the output of following program?

```
#include<stdio.h>
int main() {
    int data[] = {10, 20, 30, 40};
    int result = func(data + 1, 2);
    printf("%d\n", result);
    return 0;
}

int func(int *arr, int n) {
    int sum = 0;
    for (int i = 0; i < n; i++) {
        sum += *(arr + i);
    }
    return sum;
}
```

Handwritten notes:
- Above `int data[]`: declare?
- Next to `data + 1`: 11, 21
- Next to `2`: 1 < 2
- Next to `sum +=`: 10 + (20 + 30)
- Next to `*(arr + i)`: *(10 + 0)
- Below `*(arr + i)`: 10 +

a) 60 b) 30 c) 50 d) 40

2. Which of the following statements is false regarding the extern storage class specifier in C?

- a) It indicates that a variable or function is defined elsewhere, typically in another source file.
- ☒ b) It reserves memory for the variable or function.
- c) It can be used to declare global variables that are intended to be accessed across multiple files.
- d) It informs the linker about the existence of a variable or function defined in a different translation unit.

3. The function that calls itself is called:
a) Inline function
c) Library function
☒ b) Recursive function
d) None
4. Which of the following is NOT a pillar of OOP?
a) Encapsulation
c) Abstraction
☒ b) Inheritance
d) Compilation
5. A class is:
a) Instance of object
c) Function
☒ b) Blueprint of object
d) Method
6. Which of the following is NOT an advantage of OOP?
a) Code reusability
c) Security
☒ b) Modularity
d) High execution speed
7. IoT stands for:
a) Internet of Transfer
c) Input of Technology
☒ b) Internet of Things
d) Interface of Tools
8. Which of the following is NOT an SDLC phase?
a) System Analysis
c) Maintenance
☒ b) Testing
d) Multiprocessing
9. The process of buying and selling goods over the internet is called:
a) E-governance
c) E-learning
☒ b) E-commerce
d) E-medicine

12023

GROUP B [5×5=25]

SHORT ANSWER TYPE QUESTIONS.

10. Define Function prototype and function call with example? Write a C program using recursive function to find the sum of N natural number.
11. What is Pointer and why we use pointer in C program? Explain its advantage with simple example.

'OR'

What is UDF? Explain its types. Write a C program to find the factorial number using YRNA user define function.

[0.5

- ✓ 12. What is OOD? Suppose your friend is confused between Classes and Objects.
- a) Explain the difference using a real-life analogy (for example, "Car" as a class and "Maruti car" as an object).
 - b) Give at least two more examples to support your explanation. [1+2+2]

- ✓ 13. Kathmandu Model Secondary School wants to store all student's records, assignments, and notes on the cloud.
- a. Explain at least four advantages of using cloud storage over traditional storage.
 - b. Give three examples of a cloud service that could be used. [2+3]

- < 14. What do you mean by system testing? Differentiate between Black Box Testing and White Box Testing [1+4] >

'OR'

What is Feasibility Study? Explain its types in details with suitable example. [1+4]

GROUP 'C' [2×8=16]

LONG ANSWER TYPE QUESTIONS.

- ✓ 15. Differentiate between array and structure. Write a C program using structure to store details of 5 students and display them in descending order of marks. [3+5]
16. Write a C program to store details(id, first_name, last_name, address, salary) of N number of employees into a file "emp.txt" and display details of those employees whose address is Kathmandu . Use fprintf() and fscanf() function. [8]


"Best of Luck"



FIRST TERM EXAMINATION 2082

Class: XII 'Science'

F.M.: 75

Time: 3 hrs.

Subject: English [0041]

P.M.: 30

Candidates are required to give their answer in their own words as far as practicable. The figure in the margin indicated full marks:

Attempt all the questions.

1. Read the following passage and answer the questions that follow.

The newspapers have taken the place of the Bhagavad Gita, the Bible and the Quran with the people. For them, the printed sheet is gospel truth. The fact throws a great responsibility on the editors and the news writers.

Newspapers are a powerful influence. It is the duty of the editors to see that no false report or report likely to excite the public is published in their newspapers. The editors and their assistants have to be extra careful about the news they give and the manner in which they dress it. In a state of Independence, it is practically impossible for Governments to control the Press. It is the duty of the public to keep a strict watch on the newspapers and keep them on the right path. An enlightened public would refuse to patronize inflammatory or indecent newspapers.

Newspapers which indulge in untruth or exaggeration harm the cause they profess to espouse. I admit that there is enough untruth in enough newspapers to warrant action. But my experience is that no amount of public criticism will affect the policy of newspapers which make their livelihood by such policy... But I write this in no way to condone untruths in newspapers. I am quite clear that if newspapers weighed every word that is printed therein, we should have a speedier removal of abuses whether in the states or elsewhere.

The superficiality, the one-sidedness, the inaccuracy and often even dishonesty that have crept into modern journalism, continuously mislead honest men who want to see nothing but justice done.

The sole aim of journalism should be service. The newspaper press is a great power, but just as an unchained torrent of water submerges whole

countryside and **devastates** crops, even so, an uncontrolled pen serves but to destroy. If the control is from without, it **proves** more poisonous than want of control. It can be profitable only when exercised from within. If this line of reasoning is correct, how many of the journals in the world would stand the test? But who would stop those that are useless? And who should be the judge? The useful and the useless must, like good and evil, go on together, and man must make his choice.

A. Give short answers to the following questions in ONE SENTENCE each. [5×1=5]

- a) What have the newspapers taken the place of with the people?
- b) Who should ensure that false reports are not published?
- c) What should be the sole aim of journalism?
- d) What has crept into modern journalism?
- e) When can control of the press be profitable?

B. Substitute the underlined words in the following sentences with the suitable bold letter words from the passage. [5×1=5]

- a) The new app is very beneficial for managing daily tasks.
- b) It's my duty to take care of my pet.
- c) Her words had a big impact on my decision to study abroad.
- d) I cannot forgive cheating in any form.
- e) The wildfire destroys forests and wildlife habitats every year.

C. Fill in the blanks with suitable words from the text. [5×1=5]

- a) The printed sheet is considered as..... truth by many people.
- b) play a great role in ensuring the accuracy of the news.
- c) The duty of the is to keep a strict watch on newspapers.
- d) It is impossible for to control the press in a state of independence.
- e) The public should not support newspapers that are

2. Answer the following question in about 150 words EACH. [2×5=10]

- a) How does the poem "A Day" symbolize the cycle of life and death?
- b) How does Mrs. Baroda struggle between her emotions and the expectations society places upon her? Analyze how Chopin presents this

inner conflict and what it reveals about the position of women in the 19th century. (*A Respectable Woman*)

3. Answer the following questions in 75 words each. (Any Five) [5x2=10]

a) ✓ What does the story say about cultural differences? Explain. (*Neighbours*)

b) ✓ ~~How~~ How does the Actress manipulate the Young Woman with her words? (*A Matter of Husbands*)

OR

✓ Why did Oliver Sacks dislike school? (*On Libraries*)

c) ✓ In what ways does Sacks portray books as more than just objects? (*On Libraries*)

d) ✓ "Marriage is both a personal and social commitment." Discuss with reference to Nock's essay. (*Marriage as a Social Institution*)

e) ✓ What is the metaphorical meaning of the poem? (*A Day*)

OR

What does the phrase "a respectable woman" imply in the story? (*A Respectable Woman*)

4. Answer the following questions.

a) ✓ Write a paragraph about why cyber safety is important in today's digital world. Explain how people, especially students, can protect themselves from online scams, fake news, and data theft. [7]

b) ✓ You are living in Pokhara. The weather forecast has announced torrential rain in the hilly regions. Write a letter to your cousin in Kathmandu, advising him/her not to travel during this time. Explain the possible dangers of heavy rainfall and suggest safety precautions. [8]

c) ✓ On September 8, 2025, Gen-Z youth organized large-scale protest across Nepal, demanding transparency in governance, better employment opportunities, the lifting of the social media ban, and measures against corruption. Although the demonstration initially began peacefully, it later escalated, resulting in damage to public and private property and ultimately leading to the collapse of the government. Write a formal essay analyzing the outcomes of the Gen-Z protest, emphasizing both their positive and negative impacts in a long run. [10]

5. Rewrite the sentences as indicated in the brackets.

a) ✓ I had it repaired,? (Add a suitable tag question)

[10x1=10]

- b) All students have not passed the exam. (None/Both/Some/Most) of them have failed it too. (Choose the best option)
- c) My father spoke to me (seriously / serious / seriousness) about my negligence.
- d) Either the goats or the sheep (is/are/were/was) usually sacrificed at Dashain.
- e) He said to me, "Please help me with my homework." (Change the sentence into indirect speech)
- f) You are feeling tired. You (should / must / might / will) take some rest immediately.
- g) The man (who / to whom / whom / which) you met yesterday is my uncle.
- h) I am planning (to travel / travel / traveling) Pokhara next week
- i) She hates people making fun of her mistakes. (Change the sentence into passive voice)
- j) He offered me a cup of tea. (Change the sentence into passive voice)

6. Choose the best options:

[5x1=5]

- a) Which one of the following prefixes can be attached to the word "responsible"?
- i. ir- ii. un- iii. anti- iv. dis-
- b) Which one of the following words does not come between "book" and "boy"?
- i. boot ii. bomb iii. born iv. both
- c) The child was very timid and afraid to speak in class. The meaning of the word *timid* is:
- i. brave ii. shy iii. bold iv. strong
- d) I was under the weather yesterday, so I stayed home. The meaning of the idiom *under the weather* is:
- i. happy ii. ill iii. tired iv. angry
- e) The prefix "bi-" in the word *bicycle* means:
- i. one ii. two iii. three iv. half

Culture and tradition are the true heartbeat of our identity. Let's celebrate them in an authentic way and commit to preserving them.

Happy Tihar!

प्रथम त्रैमासिक परीक्षा - २०८२

कक्षा : १२ (विज्ञान सङ्काय)

समय : ३ घण्टा

विषय : नेपाली (००२१)

पूर्णाङ्क : ७५

उत्तीर्णाङ्क : ३०

विद्यार्थीहरूले सकेसम्म आफ्नै शब्दमा उत्तर दिनुपर्ने छ । दायाँ किनारामा दिइएको अङ्कले पूर्णाङ्क जनाउँदछ ।

सबै प्रश्नहरू अनिवार्य छन् ।

१. अनुच्छेदमा रेखाङ्कन गरिएका शब्दहरूको अक्षर संरचना र सङ्ख्यासमेत देखाउनुहोस् । (३)
जीवनमा सफल हुन चाहने व्यक्तिले मिहिनेत त गर्नुपर्छ । काम गर्दा असफल भइन्छ कि भन्ने आशङ्का गर्नुहुँदैन । संसारमा जति पनि व्यक्ति सफल भएका छन्, ती सबै लगनशील र परिश्रमी छन् ।

२. शुद्ध गरी पुनर्लेखन गर्नुहोस् । (३)

जातीए विभेद, लैङ्गिक विभेद, वर्ण वर्ग र क्षेत्रका नाममा रहेका विभेदमा पनी कमी आउँदै छ । तिन लाई पनि पूर्ण रुपमा हटाउनुपर्छ । सङ्गीताजीले भने जस्तै यशका लागि शिक्षाको खाँचो छ ।

३. कुनै एक प्रश्नको उत्तर दिनुहोस् । (२)

क) दिइएका वंशवाट एकओटा पारिभाषिक शब्द र एकओटा अनुकरणात्मक शब्द पहिचान गरी वाक्यमा प्रयोग गर्नुहोस् ।

रमेशको व्यवसायको हिसाब किताबमा बेरुजु रकमको मात्रा बढेपछि ऊ अन्योलमा पर्यो । त्यसपछि बैकठकक्षवाट जुरूक उठेर सुटुक्क घरतिर लाग्यो ।

ख) दिइएको अनुच्छेदवाट एउटा उखान र टुक्का पहिचान गरी तिनलाई वाक्यमा प्रयोग गर्नुहोस् ।

हुने विरुवाको चिल्लो पात भनेझैं सबै साथीलाई मेख मादै परीक्षामा प्राप्त गरेको श्यामको सफलतावाट उसका अभिभावक खुसीले गदगद भए । अकबरी सुनलाई कसी लाउनु पर्दैन भन्दै मेघावी विद्यार्थी श्यामले सबैको नाक राखेकोमा आफूलाई धेरै खुसी लागेको कुरा कलेजका प्रिन्सिपलले मुसुमुसु हाँस्दै सबैलाई बताए ।

४. अनुच्छेदमा रेखाङ्कन गरिएका शब्दको पदवर्ग पहिचान गरी लेख्नुहोस् : (३)

उनीहरूले असल मान्छेलाई माया गर्छन् । विद्वान् त जहाँ पनि प्रतिष्ठित हुन्छ । राम्रो काम गरेर नै नाम कमाउनुपर्छ ।

५. तलको अनुच्छेदवाट दुईओटा तत्सम र दुईओटा आगन्तुक शब्द पहिचान गरी लेख्नुहोस् : (२)

हाम्रो विद्यालयमा धेरै योग्य, असल र लगनशील विद्यार्थीहरू छन् । उनीहरू आफ्नो भविष्य निर्माण गर्न हरेक दिन ब्यागमा किताब, कापी राखेर होण्डा मोटरसाइकल र बसमा चढेर पढ्न आउँछन् ।

६. कुनै एक प्रश्नको उत्तर दिनुहोस् । (३)
- क) तलको अनुच्छेदबाट तीनबोटा उपसर्ग र तीनबोटा प्रत्यय व्युत्पन्न शब्द पहिचान गरी तिनको निर्माण प्रक्रिया देखाउनुहोस् ।
हिमाली, पहाडी वा मधेसी जो जो भए पनि तिनीहरू सबै नेपाली हुन् । हरेक नेपालीलाई सम्मान गर्न जान्नुपर्छ । दुर्व्यवहार, दुर्नाम र बदनियतले केवल बदनाम हुने गर्दछ ।
- ख) तलको अनुच्छेदबाट तीनबोटा समस्त शब्द र तीनबोटा द्वित्व शब्द पहिचान गरी समस्त शब्दको धिग्रह र द्वित्व शब्दको निर्माण प्रक्रिया देखाउनुहोस् ।
आफ्नो जन्मभूमिलाई चुपचाप विदेशीको कृपापात्र बनाउन खोज्नेहरूको देश कहिल्यै आत्मनिर्भर हुदैन । आलुफाल्नु कुरामा अनमोल समय विताउने मुलुकले अग्रगति लिंदैन । आपसी भैभगडाको जालो चटाचट काट्दै अविश्वासका खाडलहरू रातारात पुरेर जब धर्तीपुत्रहरू स्वावलम्बी बन्छन् तब जनताको भाग्योदयका होकाहरू घन्याकधुरक खुला हुन्छन् ।

७. दिइएको अनुच्छेदका वाक्यहरूलाई तृतीय पुरुषमा परिवर्तन गरी पुनर्लेखन गर्नुहोस् । (४)
- तिमी असल छौ । तिमी आफूभन्दा ठुलालाई आदर गर्छौ । तिमीलाई सबैले सम्मान गर्छन् । तिमी भविष्य उज्ज्वल छ ।

८. दिइएको अनुच्छेदलाई भूतकालमा परिवर्तन गर्नुहोस् । (४)
- सम्राट स्कूल जान्छ । सम्राटले मिहिनेत गरी पढ्छ । सम्राटले आफ्नी बहिनी सम्पदालाई पनि पढ्न सिकाउँछ । सम्राट आज्ञाकारी छ ।

९. तलका चार बोटा जटिल (मिश्र वा संयुक्त) वाक्यहरूलाई आठ बोटा सरल वाक्यमा छुट्याउनुहोस् । (४)
- वसन्त ऋतु सुरु हुन्छ र चारैतिर फूलहरू फुल्ले बाल्छन् । जहाँ फूल फुल्छ त्यहाँ वासना चल्छ । फूलको सुवास मिठो हुन्छ र यसको सुवासमा सबै मुग्ध हुन्छन् । मलाई फूलको वासना मन पर्छ र म सधैं बगैँचामा भुल्छु ।

१०. तलको अनुच्छेद पढी सोधिएका प्रश्नको उत्तर दिनुहोस् । (५)
- जीवनको एउटा तर महत्त्वपूर्ण अंशलाई समेट्ने विशिष्ट घटनाको सिलसिलेवार प्रस्तुति कथा हो । यिनै घटनाको कार्यकारण शृङ्खलालाई कथानक भनिन्छ । कथाको आकार र स्वरूप तयार पार्ने कार्यमा कथानकले महत्त्वपूर्ण भूमिका निर्वाह गर्ने भएकाले यसलाई कथाको मुख्य तत्त्वका रूपमा लिने गरिन्छ । कथामा घटना अघि बढाउने वा कार्यव्यापार सम्पन्न गर्न पात्रको प्रयोग गरिएको हुन्छ । पात्र मानवीय र मानवेतर गरी दुवै किसिमका हुन्छन् । कथा भन्ने पात्रलाई समाख्याता भनिन्छ । समाख्याता कथा भित्रकै पात्र पनि हुन सक्छ र कथा बाहिरको पनि हुन सक्छ । कथाको पात्र आफैले कथाको समाख्यान गरेको छ भने त्यसलाई प्रथम पुरुष दृष्टिविन्दु भएको कथा र अन्य पात्रका माध्यमबाट कथाको समाख्यान गरिएको छ भने त्यसलाई तृतीय पुरुष दृष्टिविन्दुको प्रयोग भएको कथा भनिन्छ । प्रथम पुरुष दृष्टिविन्दु भएका कथामा म र हामी जस्ता प्रथम पुरुषलाई बुझाउने पदको प्रयोग गरिएको हुन्छ भने तृतीय पुरुष दृष्टिविन्दु भएका कथामा त्यो, ऊ तिनीहरू आदि तेस्रो पुरुष बुझाउने पदको प्रयोग गरिएको हुन्छ । घटनाको समय, स्थान र त्यसबाट उत्पन्न हुने वातावरणलाई परिवेश भनिन्छ । कथाले भन्न खोजेको मूल कुरा सारबस्

हो । यसैलाई सन्देश पनि भनिन्छ । यी सबैलाई प्रस्तुत गर्ने माध्यम चाहिँ भाषा हो । कथामा गद्य भाषाको प्रयोग हुन्छ । कथालाई प्रभावकारी रूपमा अधि बढाउन पात्रका विचमा संवाद पनि हुन सक्छ । कथाको भाषालाई वर्णनात्मक, पत्रात्मक, संस्मरणात्मक, नाटकीय, आदि शैलीमा प्रस्तुत गर्न सकिन्छ । यसैले कथानक, पात्र, परिवेश, दृष्टिबिन्दु, भाषाशैली र सारवस्तु यी सबै कथाका तत्व हुन् ।

प्रश्नहरू :

- अ) कथा भनेको के हो ?
- आ) कथामा कति किसिमका पात्र हुन्छन् ?
- इ) तृतीय पुरुष दृष्टिबिन्दु भएका कथामा कस्ता पदको प्रयोग गरिएको हुन्छ ?
- ई) कथाको भाषालाई कुन शैलीमा प्रस्तुत गर्न सकिन्छ ?
- उ) कथाका तत्व के के हुन् ?

११. तलको अनुच्छेदबाट मुख्य मुख्य चार बुँदा टिपी एक तृतीयांशमा सारांश लेख्नुहोस् । (५)

सहकार्यबाट व्यक्तिलाई आत्मनिर्भर बनाउने र समुदायको दिगो विकास गर्ने उद्देश्यले सहकारीको अवधारणा आएको हो । स्तन्त्रताका लागि स्वावलम्बन, समुन्नतिका लागि पारस्परिकता र दुवैका लागि सहकारिता सहकारीको नारा रहेको छ । सहकार्य गर्दा आम जनसमूहमाभ छरिएर रहेको पुँजी, प्रतिभा र प्रविधिको एकीकृत उपयोग भई यसबाट आर्थिक उन्नति हुने भएकाले व्यक्ति आत्मनिर्भर बन्नका साथै समाजको उन्नति हुन्छ । सहकारीमा सहकार्यलाई महत्त्व दिने भएकाले यसमा न्यून पुँजी भएका व्यक्ति पनि आवद्ध हुन सक्छन् । न्यून पुँजी भएका व्यक्तिलाई एक थुकी सुकी सय थुकी नदी भने भई यसमा आवद्ध भएर एक एक थोपा जम्मा गरी गागो गर्ने सुविधा छ । सबै किसिमका व्यक्तिलाई समेट्न सक्ने हुँदा अधिकसित, विकासोन्मुख र विकसित सबै देशमा सहकारी खुलेका छन् । आर्थिक स्थिति कमजोर भएका व्यक्तिलाई पनि सहकारीले प्राथमिकतामा राख्ने भएकाले समानुपातिक आर्थिक समुन्नतिका लागि नेपाल जस्तो विकासोन्मुख देशका लागि सहकारी जस्तो साझा संस्था निकै उपयोगी छ ।

१२. कुनै एक प्रश्नको उत्तर दिनुहोस् । (४)

क) साझा प्रकाशन ललितपुरले त्रिवि पुस्तकालय काठमाडौँलाई एक चिहान उपन्यास बुझाई भुक्तानी लिए वापत गरिदिएको भरपाईको नमुना लेख्नुहोस् ।

ख) खेलकुद कार्यक्रममा सहभागी हुन पाऊँ भनी आफ्ना प्रधानाध्यापकलाई एउटा निवेदन लेख्नुहोस् ।

१३. कुनै एक प्रश्नको उत्तर दिनुहोस् । (५)

क) आफ्नो विद्यालयको वार्षिकोत्सव कार्यक्रमको विषयलाई समेटी एउटा प्रतिवेदन तयार पार्नुहोस् ।

ख) 'पेट्रोलियम पदार्थको मूल्यवृद्धि' ले पारेको नकारात्मक असरका बारेमा १५० शब्द सम्ममा टिप्पणी लेख्नुहोस् ।

१४. कुनै एकको व्याख्या गर्नुहोस् । (४)

स्क) प्रकृतिसित जुध्नु नै जीवनको रमाइलो हो भन्ने भावनालाई ह्याकुलाले छोपेर नै म सदूरपश्चिमको भ्रमणमा निस्केको थिएँ ।

(ख) त्यो आउँदा तिमी पहिले त सपना हो कि
भनेर छामछुम गर्ने छौ तर त्यो हिउँ र आगोभन्दा पनि
बढ्ता छोड्ने भएर आउँछ

१५. कुनै एक प्रश्नको उत्तर दिनुहोस् ।

(४)

(क) दिइएको कथांश पढी सोधिएका प्रश्नको उत्तर लेख्नुहोस् :

कालोनिलो आकाशमा उत्तरतिर ध्रुवतारा त्यसको दाहिनेतिर साततारा र देखेतिर पाँच ताराका
उज्याला तारामण्डलले तिनलाई मोहित तुल्यायो । पूर्वदेखि पश्चिमतिर केही दक्षिणतिर
ठल्किएको आकाशगङ्गा दमयन्तीका दृष्टिमा सङ्गीतमय हुन थाल्यो । आकाश र पृथ्वी यस्ता
सुन्दर छन् । तिनका बिच आफू एकली विरहिणी अवस्थामा अनुभव गरेर दमयन्ती कोमल
मधुर स्वरमा वियोग र पीडाको गीत गाउन थालिन् । कतिबेरसम्म तिनले गीत गाइन्, तिनले
आफै थाहा पाइन् । आकाशमा विस्तारै विस्तारै उम्रिएर आउँदै पातला-बादलले त्यो सुन्दर
आकाशलाई दमयन्तीका हृदयमा धूमिल तुल्याउँदै लग्यो ।

प्रश्नहरू :

(अ) माथिको अनुच्छेदमा कस्तो परिवेशको चित्रण गरिएको छ ?

(आ) आकाश र दमयन्तीको हृदयका बिच के समानता छ ?

(ख) तलको कवितांश पढी दिइएका प्रश्नको उत्तर लेख्नुहोस् ।

त्यो हुरी भएर आउने छ

तिमी पात भएर पछ्याउने छौ

उहिले त्यो जीवनलोकबाट भरेर

जूनजस्तै पोखिँदा सारा जडता सगबगाएको थियो बा ।

प्रश्नहरू :

(अ) 'तिमी पात भएर पछ्याउने छौ' भन्नुको तात्पर्य के हो ?

(आ) यहाँ प्रस्तुत जडताले के कुरालाई सङ्केत गरेको छ ?

६. कुनै एक प्रश्नको प्रतिक्रियात्मक उत्तर दिनुहोस् ।

(४)

(क) घनघस्याको उकालो नियात्रामा भनिए जस्तै घनघस्या क्षेत्रको बाटाको यात्रा आजसम्म पनि
स्थितै कठिन होला त ? आफ्ना प्रतिक्रिया प्रस्तुत गर्नुहोस् ।

(ख) आफ्नो दृढ सङ्कल्प पूरा गर्न मानिस जतिसकै सङ्घर्ष गर्न पनि तयार हुन्छ भन्ने भनाइप्रति
तपाईंको के प्रतिक्रिया छ ? विरहिणी दमयन्ती कथाका आधारमा पुष्टि गर्नुहोस् ।

७. कुनै एक प्रश्नको समीक्षात्मक उत्तर दिनुहोस् ।

(८)

(क) 'एक चिहान' उपन्यासले सामाजिक कुसंस्कारका बारेमा कस्तो दृष्टिकोण प्रस्तुत गरेको छ ?
लेख्नुहोस् ।

(ख) आमाको सपना के हो र त्यो कसरी पूरा हुन सक्छ, 'आमाको सपना' कविताका आधारमा
चर्चा गर्नुहोस् ।

कुनै एक शीर्षकमा कमिमा २५० शब्दसम्मको निबन्ध लेख्नुहोस् ।

(८)

(क) मेरो देश : मेरो गौरव

(ख) जेष्ठ नागरिकप्रति हाम्रो दायित्व

(ग) स्वरोजगार

सफलताको शुभकामना !!!